

Practical No. 2: Write an X86/64 ALP to accept a string and to display its length

```
section .data
msg1 db 10,13,"Enter a string:"
len1 equ $-msg1
section .bss
str1 resb 200 ;string declaration
result resb 16
section .text
global _start
_start:
;display
mov Rax,1
mov Rdi,1
mov Rsi,msg1
mov Rdx,len1
syscall
;store string
mov rax,0
mov rdi,0
mov rsi,str1
mov rdx,200
syscall
call display
;exit system call
mov Rax ,60
mov Rdi,0
syscall
%macro dispmsg 2
mov Rax,1
mov Rdi,1
mov rsi,%1
mov rdx,%2
syscall
%endmacro
display:
mov rbx,rax ; store no in rbx
mov rdi,result ;point rdi to result variable
mov cx,16 ;load count of rotation in cl
up1:
rol rbx,04 ;rotate no of left by four bits
mov al,bl ; move lower byte in al
and al,0fh ;get only LSB
cmp al,09h ;compare with 39h
jg add_37 ;if greater than 39h skip add 37
add al,30h
jmp skip ;else add 30
add_37:
add al,37h
skip:
mov [rdi],al ;store ascii code in result variable
inc rdi ; point to next byte
```

```
dec cx ; decrement counter
jnz up1 ; if not zero jump to repeat
dispmsg result,16 ;call to macro
ret
```